Final Project Proposal
Group 8, Station 6
Lab B01

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Overview

Our chosen converter is the **Single-Phase Standalone Inverter**. It takes a fixed DC input and supplies a resistive load with a regulated 60 Hz sinusoidal voltage.

Operating envelope and points

Instrument	Voltage [V]	Current [A]	Power [W]	Envelope
ITECH IT-M3632 DC supply	60	30	800	58V
SEMITEACH IGBT	400 AC	50	-	7.9A
47 µH Inductor	250	19.3	-	3.7A
Load (12.5Ω)	-	-	-	12.5Ω - 250Ω

Quantity	Operating Point 1	Operating Point 2
Output voltage	$35.4V_{pk}$ (36%)	$42.4V_{pk}$ (23%)
Input voltage	50V (13%)	50V (11%)
Inductor current RMS	1.7A – 2.3A (88%)	2.7A – 3.0A (84%)
Inductor current ripple	4.1A – 5.5A (83%)	5.4A – 6.7A (79%)
Output power	25W (97%)	72W (90%)
Load resistance	25Ω	12.5Ω

Binding constraint: The 58V DC limit, caps the output at about $39V_{rms}$

Margin = (limit - value) / limit

Steady-state targets

Measurable Value	Instrument	Operating Point 1	Operating Point 2
Load Voltage	Probe 1	$25V_{rms}$, $35.4V_{pk}$ 60Hz	$30V_{rms}$, $42.4V_{pk}$ 60Hz
Load Current	Probe 2	$1A_{rms}$	$2.4A_{rms}$
Load Power	Fluke39	25W	72W
Current Ripple	Oscilloscope Measure	$5.5A_{pp}$ – $8.3A_{pp}$	$5.5A_{pp}$ – $8.3A_{pp}$
Inductor Current	Fluke39	$1.04A_{rms}$	2.43_{rms}
Capacitor Current	Fluke39	$0.29A_{rms}$	$0.35A_{rms}$

Preliminary switching frequency

At f_{sw} = 10kHz, the 20 kHz ripple is about 6.8x above the 2.95 kHz resonance and losses are small at 50V. We want to achieve a 60Hz sinusoidal AC wave with a small inductor. A unipolar gate pattern, we will be able to get inductor current ripple to be approximately 20% (6.7 / 32.3 ≈ 21%).

Sensing and protection

Signal	Probe	Probe scaling	Overvoltage & Overcurrent Threshold	ADC OV and OC Threshold Reading	Firmware Action
Output Voltage	VP1	10:1	±50 V	0.602 V, 2.732 V	Turn the simulation off
Input Voltage	VP2	10:1	55 V	2.839 V	Turn the simulation off
Output Current	CP1	100mV/A	-	-	-
Inductor Current	CP2	100mV/A	±9A	1.528 V, 1.806 V	Turn the simulation off

Connection diagram

See Figure 1

Safety plan

Startup

1. Ensure the station instruments are off before operating our own procedures
2. Configure the ITECH by putting it to source mode (50V, 10A, 60V OVP) and leave its output off
3. Setup oscilloscope probes and Fluke39 measurement points
4. Set our resistance bank to the appropriate value
5. Check wiring against our diagram with all group members tracing each connection. Get a check from a TA or professor to further ensure proper connections
6. Power the launchpad and flash the PLECS file

Normal Shutdown

1. Leave the load connected and use the enable switch in PLECS to disable control
2. Ramp the ITECH to zero, ensuring the display is below 5V
3. Turn the power supply off, then disconnect and return cables

Fault Response

1. Turn ITECH output off
2. Keep note of values when the fault happens
3. Confirm the display is below 5V on the ITECH display and with a multimeter on the DC link and capacitors before handling
4. Investigate with TA, professor and group members before powering on again

Work plan

Task	Lead	Support	Target Date
Switched PLECS station model (full bridge, LC filter, load)	Jemmson	Albert	Oct 20
HIL model parameter and I/O checks; first controller communication	Albert	Wajeeh	Oct 23
Connection diagram finalized	Temuulen	Jemmson	Oct 27
Design review: plant, compensator, dynamic targets	Wajeeh	Temuulen	Bench 1, Oct 30
Firmware (PLECS Coder) and HIL checklist	All	All	Nov 13
Bench 2: HIL sign-off, closed loop at reduced power	All	All	Fri Nov 20
Bench 3: captures, ITECH and THD readings, steps, log	All	All	Fri Dec 4
Report: targets, hardware/control, HIL, results	All	All	Fri Dec 11

AI disclosure

Claude (Anthropic) was used to draft and structure this document. It suggested operating points, protection settings, envelope, ripple, and ADC-scaling calculations.

Figures

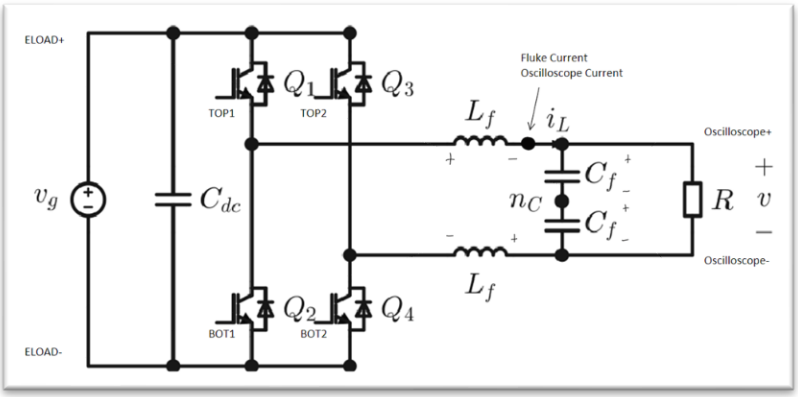


Figure 1: Wiring Diagram